Fiscal Year:	FY 2016	**Task Last Updated:**	FY 02/07/2017
PI Name:	Baulch, Janet E. Ph.D.		
Project Title:	High LET Radiation Induced Carcinogenesis: Epigenetic Mechanisms in Mouse Models of Human Cancers		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	NOTE: PI previously at University of Maryland; moved to UCA Irvine in summer 2013 (Ed., 11/11/13)		
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Space Radiobiology NNJ11ZSA001N
Start Date:	08/01/2013	**End Date:**	07/31/2016
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: Extended to 7/31/2016 per NSSC information (Ed., 7/14/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX13AK69G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	[Editor's note 11/11/2013. Note this is a continuation of "High LET Radiation Induced Carcinogenesis: Epigenetic Mechanisms in Mouse Models of Human Cancers" (grant NNX12AE79G); see that project for previous reports.] This was a collaborative project with other NASA investigators who are characterizing mouse models of human lung and colon cancers. Data from both our laboratory and the literature suggest that epigenetics play a significant role in carcinogenic processes. By describing the changes in DNA methylation and miR regulation of gene expression for these mice, we planned to provide NASA with mechanistic data regarding carcinogenic initiation and progression. By using normal, cancer, and metastatic cancer cell lines we will obtain information regarding epigenetic changes that occur during carcinogenesis based on stage. For example, loss of control of gene expression for extracellular matrix (ECM) remodeling genes may foster metastatic behavior in colon cancer. For this reason we will include ECM gene promoter DNA methylation and gene expression among the loci evaluated in this study to the hypothesis that dis-regulation of epigenetic marks and gene expression changes with disease progression. Similarly, the work of other NASA investigators suggests that we will see epigenetically regulated differences among cell lines in expression of inflammatory response pathway gene regulation among lung cell lines. These experiments will provide information regarding radiation responses of these mouse tissues and human cell lines based on the LET (linear energy transfer) of the radiation used. Abnormal epigenetic silencing of genes can occur at a variety of times during tumor progression. If so, does radiation exposure push normal cells to a cancer phenotype and cancer cells to a metastatic phenotype? This question is of importance to NASA since astronauts are of an age where they likely go into missions with precursor lesions. It is important to understand whether their occupational radiation exposures lead to increased cancer risk. Tasks: 1. Evaluate normal and cancer cell lines from lung and colon for responses to low dose high LET radiation exposure. 2. Establish epigenetic profiles for in vivo radiation induced cancers in mouse models of human lung and colon cancers.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Reducing the uncertainties associated with cancer risk models for radiation exposures involved in space flight is of considerable importance in light of long-term occupation of the International Space Station (ISS) and other future manned missions by astronauts. A greater understanding of these carcinogenic mechanisms also benefits Earth-bound populations. Colon cancer is the third most common cancer. Precancerous lesions in the GI tract are found in approximately 1 out of every 10 people by age 40 and this incidence increases significantly with age. In addition to being the largest target for radiation effects, lung cancer is the most prevalent form of cancer in men and women. Each of these cancers has hallmark mutations associated with them. On top of these genetic changes, our understanding of carcinogenesis is complicated by epigenetic alterations to the genome. These epigenetic changes result in misregulation of gene expression for pathways such as cell cycle regulation, DNA repair, apoptosis, and extracellular matrix remodeling. A better understanding of these epigenetic mechanisms may lead to improvements in clinical treatments for lung, colon, and other cancers.
Task Progress:	Aim 1 will use normal, cancer, and metastatic cancer cell lines to test the hypotheses that functional differences in profiles of DNA methylation and miR mediated gene regulation are I) associated with tumor aggressivity, II) altered following radiation exposure to affect epigenetic carcinogenic progression, and III) differ based on response to acute versus fractionated 56Fe ion irradiation. Aim 2 will test three hypotheses in mouse models of human lung and colon cancer. I) miR expression affects carcinogenic initiation and progression by altering target gene expression, II) aberrations in gene promoter DNA methylation affect changes in gene expression contributing to the initiation/perpetuation of radiation carcinogenesis, III) hypomethylation of repeat elements allows reactivation of the elements that may increase the likelihood of retrotransposition and genomic instability. The project resulted in three publications addressing the radiation response in the context of cancer treatment or ameliorating the effects of radiation exposure. In the publication by Thomas and colleagues (2013) we demonstrated a, then novel but now commonly accepted, mechanism for radioresistance in breast cancer—exosome mediated paracrine signaling. Exosomes are nm sized microvesicles secreted by one set of cells that can effect change in target cell phenotype when exosome cargo is released into those target cells. These functionally relevant cargo include miR and other coding and non-coding RNA, proteins, lipids, and other effector molecules. This publication demonstrated that exosomes could protect cancer cells from radiotherapy, decreasing treatment efficacy. We extended this work to develop the hypothesis of ‘bad’ vs. ‘good’ exosomes in the Baulch et al., 2016. While the study by Thomas and colleagues defined a ‘bad’ exosome, in the 2016 project we determined that human neural stem cell-derived exosomes were ‘good’ exosomes that could mitigate the effects of radiation exposure on cognition as well as neuronal structure and radiation induced neuroinflammation. Subsequent yet unpublished follow up studies suggest that among the functionally relevant cargo of those stem cell derived exosomes are miR that have gene targets relevant to the molecular, structural and behavioral improvements observed in the irradiated animal’s brain. We also published a study of dose per pulse and dose rate effect on cancer cell survival (Lasio et al., 2014). Further, in conjunction with another NASA funded project, 6 other publications addressing the role for epigenetic mechanisms in the high and low LET radiation responses. The studies related to this project have been presented at the 2014-2016 NASA Human Research Program (HRP)/Space Radiation Investigators’ Workshops in Galveston, the 13th International Workshop on Radiation Damage to DNA, the International Congress for Radiation Research in Kyoto, Japan (2015). Additionally, work has been presented at the 2014 and 2015 Environmental Mutagenesis and Genomics Society meetings (Posters) and at the 2014-2016 Radiation Research Society meetings (various Oral and Poster presentations). Publications (see also Cumulative Bibliography): Thomas SN, Liao Z, Clark D, Chen Y, Samadani R, Mao L, Ann DK, Baulch JE, Shapiro P, Yang AJ. "Exosomal proteome profiling: a potential multi-marker cellular phenotyping tool to characterize hypoxia-induced radiation resistance in breast cancer." Proteomes. 2013 Sep 1;1(2):87-108. http://dx.doi.org/ ; PubMed PMID: 24860738 ; PubMed Central PMCID: PMC4029595 Lasio G, Guerrero M, Goetz W, Lima F, Baulch JE. "Effect of varying dose-per-pulse and average dose rate in X-ray beam

	irradiation on cultured cell survival." Radiat Environ Biophys. 2014 Nov;53(4):671-6. http://dx.doi.org/ ; PubMed PMID: 25169705 Baulch JE, Acharya MM, Allen BD, Ru N, Chmielewski NN, Martirosian V, Giedzinski E, Syage A, Park AL, Benke SN, Parihar VK, Limoli CL. "Cranial grafting of stem cell-derived microvesicles improves cognition and reduces neuropathology in the irradiated brain." Proc Natl Acad Sci U S A. 2016 Apr 26;113(17):4836-41. http://dx.doi.org/ ; PubMed PMID: 27044087; PubMed Central PMCID: PMC4855546 Presentations from 2016-2017: October 2016 Radiation Research Society meeting in Kona HI. Characterizing the Functional Cargo of Human Neural Stem Cell-Derived Microvesicles that Mitigate the Effects of Cranial Irradiation. Ning Ru, Liping Yu, Erich Giedzinski, Munjal M. Acharya, Janet E. Baulch. February 2016 HRP meeting in Galveston TX. MICROVESICLE-MEDIATED SIGNALING IN THE BRAIN'S RADIATION RESPONSE. N. Ru, L. Yu, A. Syage, M. Le, C.L. Limoli, J.E. Baulch.
Bibliography Type:	Description: (Last Updated: 05/01/2017)
Articles in Peer-reviewed Journals	Baulch JE, Acharya MM, Allen BD, Ru N, Chmielewski NN, Martirosian V, Giedzinski E, Syage A, Park AL, Benke SN, Parihar VK, Limoli CL. "Cranial grafting of stem cell-derived microvesicles improves cognition and reduces neuropathology in the irradiated brain." Proc Natl Acad Sci U S A. 2016 Apr 26;113(17):4836-41. http://dx.doi.org/10.1073/pnas.1521668113 ; PubMed PMID: 27044087; PubMed Central PMCID: PMC4855546 , Apr-2016
Articles in Peer-reviewed Journals	Acharya MM, Baddour AAD, Kawashita T, Allen BD, Syage AR, Nguyen TH, Yoon N, Giedzinski E, Yu L, Parihar VK, Baulch JE. "Epigenetic determinants of space radiation-induced cognitive dysfunction." Scientific Reports. 2017 Feb 21;7:42885. http://dx.doi.org/10.1038/srep42885 ; PubMed PMID: 28220892; PubMed Central PMCID: PMC5318883 , Feb-2017